

Weight and flight assessment

Revision 2 / Read before committing to the complete build

Flight and hand launch are not cleared. The calculations below screen the design. They do not replace airfoil/propeller data or tests of the built aircraft.

Calculated item	Result and limit
Nominal all-up mass	2.63 kg using sliced object mass and stated component/stock allowances. Actual glue, hardware and material can change this.
Wing geometry	0.30445 m ² exposed planform; 169.83 mm mean aerodynamic chord; aspect ratio 13.14.
Balance reach	Battery centre required for geometric quarter-MAC: Y=-33.7 mm; available Y=-60..+20 mm. Quarter-MAC is 522.3 mm aft of nose; this is not an approved flight CG.
Tail geometry	Approximate V-tail pitch area 0.0501 m ² , 35-degree cant and 0.368 m moment arm. Horizontal tail volume about 0.357.
Stability screening	A linear sensitivity study at quarter-MAC spans about -1% to +21% static margin under the stated uncertain tail/downwash/body assumptions. This does not establish a safe CG range.

Speed matters

All-up mass	Stall-speed sensitivity	1.3 times stall
2.60 kg	10.7-13.1 m/s	13.9-17.0 m/s
2.85 kg	11.2-13.7 m/s	14.5-17.8 m/s
3.15 kg	11.7-14.4 m/s	15.3-18.7 m/s

Assumes sea-level density 1.225 kg/m³ and maximum lift coefficient 0.8-1.2. Those lift coefficients are scenarios, not measured values. Lower air density increases the required true airspeed. The 1.3 multiplier is a screening reference, not a launch-speed prescription.

An ordinary hand throw is not established as adequate. The launch plan needs a measured weight, demonstrated propulsion performance and an experienced fixed-wing review. Do not treat the autopilot launch-trigger speed as aerodynamic flying speed.

Filament and propulsion

Print estimate	Current generic slicing run
Installed pieces	42 pieces; approximately 728 g of object extrusion.
Filament with supports	1.30 kg; approximately 1.49 kg with 15% waste allowance. Budget two 1 kg spools.
Slicing basis	PrusaSlicer 2.9.6; 0.4 mm nozzle, 0.2 mm layers, shells 2 walls/5%, controls 3 walls/10%, mounts 4 walls/30%, 4 top/bottom layers, gyroid, automatic supports; PLA density 1.24 g/cm ³ .
Workshop action	Reorient shells to reduce support and reslice for the actual printer. Record each finished mass. These settings are an estimate, not a strength-qualified recipe.

Selected propulsion: Emax ECO II 2807 1300KV, Gemfan 9045CL three-blade, 4S pack and 40 A ESC. No matching verified thrust/current map was found, so thrust capability has not passed.

The drag/power sensitivity calculation gives roughly 4-19 A at 16-20 m/s, using assumed drag coefficients 0.035-0.070, span efficiency 0.65-0.85, combined propulsion efficiency 0.45-0.65, 14 V under load and 10 W avionics. This broad range is not a current prediction; some speed/lift combinations may be near stall.

Flight-time planning: 12-20 minutes only if average total current is 12-20 A and 20% of the nominal capacity remains at landing. Time = 249.6 / average amperes, in minutes. Replace assumed current with a calibrated flight log.

What must close before flight

Measure current, static thrust, voltage sag and motor/ESC temperature with the actual propeller. Stay within the lowest continuous rating; do not use an unmeasured 40 A assumption as a pass. Static thrust alone cannot establish launch acceleration or thrust at flying speed.

MG90S load screening uses its 1.8 kgf-cm stall rating at 4.8 V and a 50% working allowance. With 10 mm servo arms, 15 mm horns and an assumed hinge coefficient of 0.2 at 20 m/s, calculated servo torque is 0.043 Nm for an aileron and 0.014 Nm for a tail control, below the 0.088 Nm allowance. This coefficient is unmeasured; it does not approve a speed or control limit. Verify CG, trim and combined V-tail travel. Prove the wing/joiner, motor mount and landing-gear attachments under agreed loads. Record radio failsafes, range, control direction, sensor calibration and actual balance using the Electronics Team chapter.

Calculation basis: NASA Glenn lift/drag equations and induced-drag relation. Tail planform was projected for aerodynamic estimation only; no projected outline was used as a foam cutting pattern.

NASA: induced drag | NASA: drag equation